MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFSC200 – BIOMETRICS AND IMPRESSION ANALYSIS

(2024 ADMISSION ONWARDS)

Duration:1 hour 30 minutes Maximum Marks: 50

Part A – Short Answers

1. Patent fingerprint – visible, formed by blood/paint/ink.

2. Chloride ions react with silver nitrate.

3. Fluorescent powders: better visibility, contrast under ALS.

4. Toe alignment helps identification and gait analysis.

5. Fingerprinting unconscious persons: rolled method, assisted support.

6. Ridge detail: ridge endings, bifurcations, unique morphology.

7. Lifting smooth surface impressions: tape lift, gel lifters.

8. Palm prints: large area, unique ridge patterns.

9. Comparison procedure: side-by-side, ridge matching, magnification.

10. Patent fingerprint: visible due to foreign material.

11. Ninhydrin reacts with amino acids.

12. Human vs. animal footprints: gait type, claw presence, toe numbers.

Part B – Short Essay

Evaluation:
4 marks – 4 points well-explained
3 marks – 3 good points
2 marks – partial answer
1 mark – minimal content

13. Challenges with dead body fingerprints: skin slippage, decomposition, rehydration needs. (Any valid 2 points with good explanation may provide full marks)

14. Lifting developed prints: adhesive tape, gel lifters, electrostatic lifter.

15. Lip prints – physical (powders, tapes) and chemical (Sudan dyes, silver nitrate).

Physical:
- Powder dusting
- Magna brush
- Adhesive tape

Chemical:
- Sudan dyes
- Silver nitrate
- Iodine fuming

16. Dry surface lifting: electrostatic lifting, gel lifters, transparent tape.

17. Filters in forensic photography: UV, IR, polarizing filters.

Points:
- UV filters
- Polarizing filters
- Infrared filters
- Used for contrast enhancement & reducing glare

18. Foot impression development: chemical enhancement, powdering, casting.

Points:
- Chemical enhancement (Any)
- Powder dusting
- Casting materials (dental stone)
- Electrostatic lifting

19. Importance of fingerprint types: classification, identification value.

Points:
- Loop, whorl, arch classification
- Helps in narrowing suspects
- Permanent & unique patterns
- Basis for comparison

Part C- Essay type

Evaluation:
10–12 strong points → 10 marks
8–9 points → 8 marks
6–7 points → 6 marks
<5 points → low marks

20. Footprint sequence reconstruction: gait, stride, direction, height estimation, sequence order, suspect linkage.

21. ALS, oblique lighting, filters: UV/IR detection, contrast enhancement, barrier filters, photographic documentation.